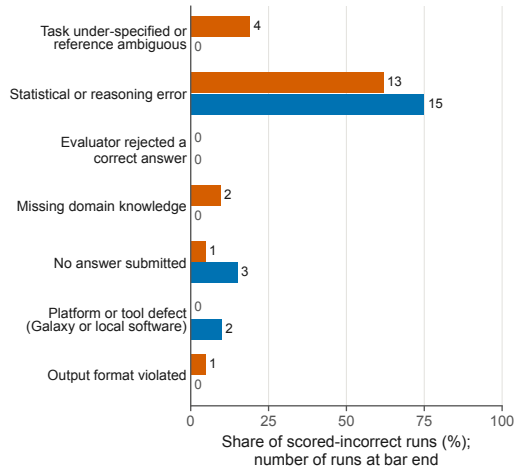


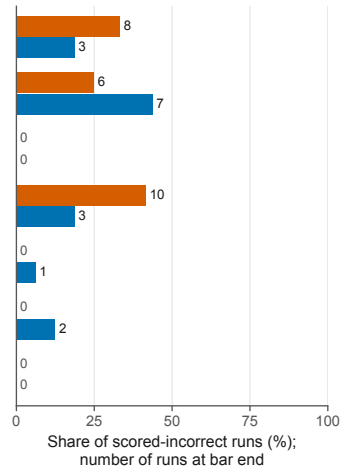
a One of three replicate runs scored incorrect: mostly statistical or reasoning errors

Open-ended code: 21 runs in 21 replicate sets
Galaxy: 20 runs in 20 replicate sets



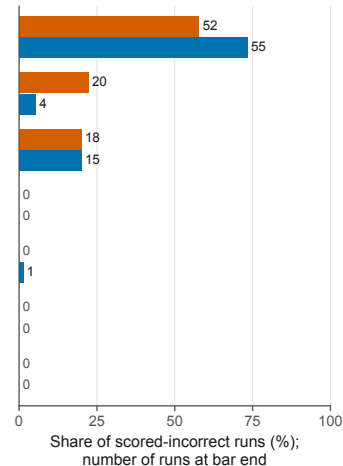
b Two of three replicate runs scored incorrect: causes were mixed

Open-ended code: 24 runs in 12 replicate sets
Galaxy: 16 runs in 8 replicate sets



c All three replicate runs scored incorrect: mostly under-specified tasks or evaluator rejections

Open-ended code: 90 runs in 30 replicate sets
Galaxy: 75 runs in 25 replicate sets



Open-ended code condition
Galaxy condition

BixBench-Verified-50, all five model configurations pooled (250 replicate sets per condition, as in Fig. 4b). Every scored-incorrect run (135 open-ended code, 111 Galaxy) carries one primary cause from the failure ledger (Supplementary Table 14e). Replicate sets with no scored-incorrect run are not shown.